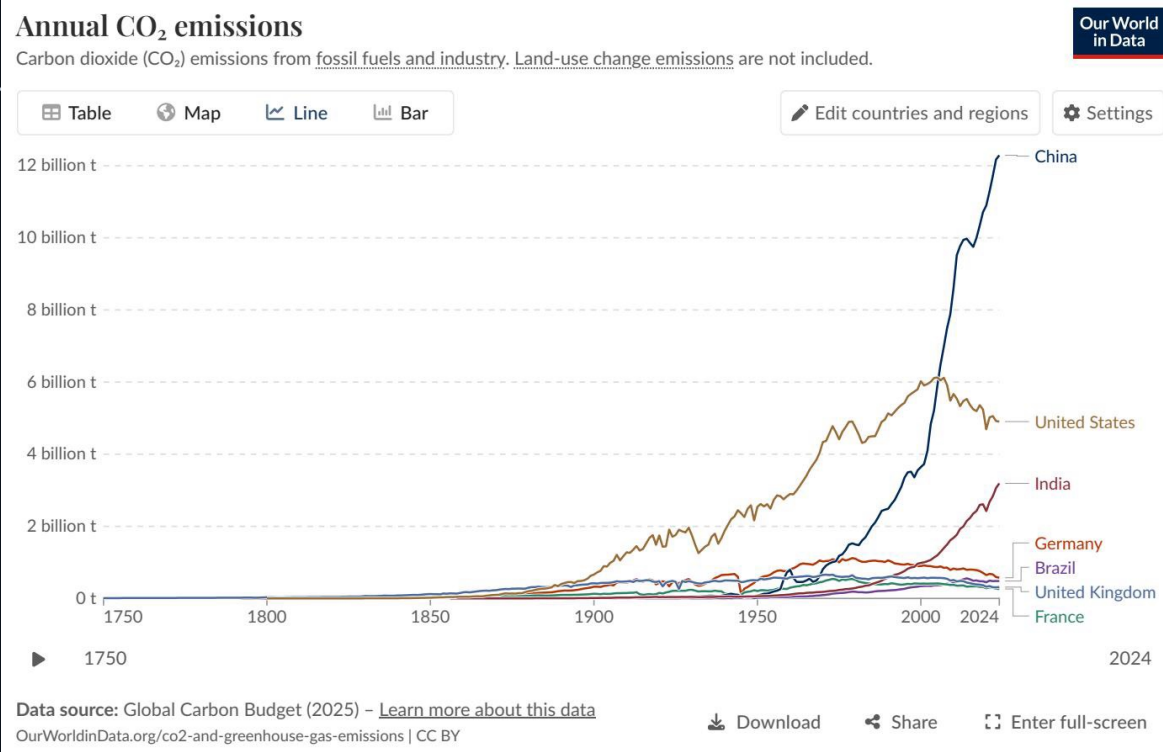


Global CO₂ Emissions: Trends, Regional Shifts, and Per Capita Ineq

Scope: 1750–2024 | Audience: policy analysts & climate negotiators



Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

Mauna Loa (NOAA), February 2026

429 ppm

~280 ppm pre-industrial baseline → >50% increase

A rise that once took thousands of years is occurring in ~2 centuries.

Multiple measurement systems confirm acceleration since 1958.

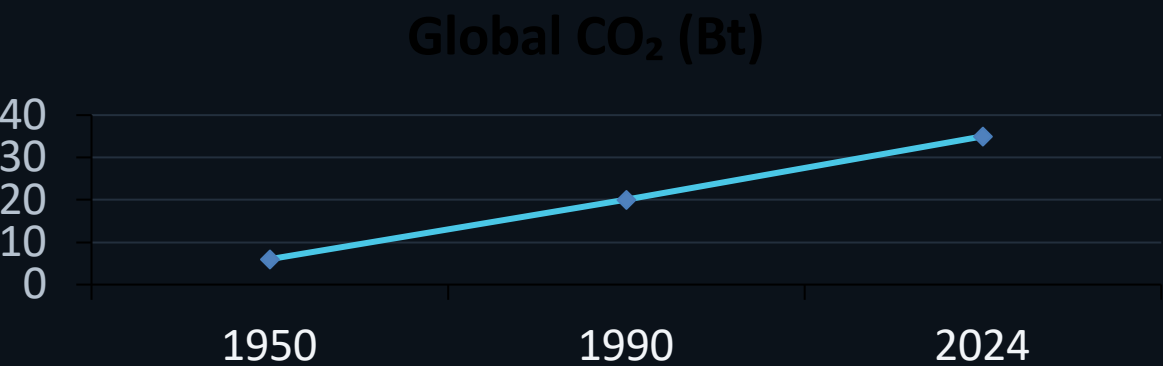
Primary driver: combustion of coal, oil, and natural gas.

Context

1958: start of Mauna Loa record
~280 ppm: pre-industrial baseline
2026: 429 ppm measured

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Bi

1950: ~6 billion tonnes (Bt) CO₂
1990: >20 Bt (nearly quadrupled)
2024: >35 Bt from fossil fuels and industry
Growth has slowed, but global emissions have not yet peaked.



Source: Global Carbon Budget 2025 (via Our World in Data)

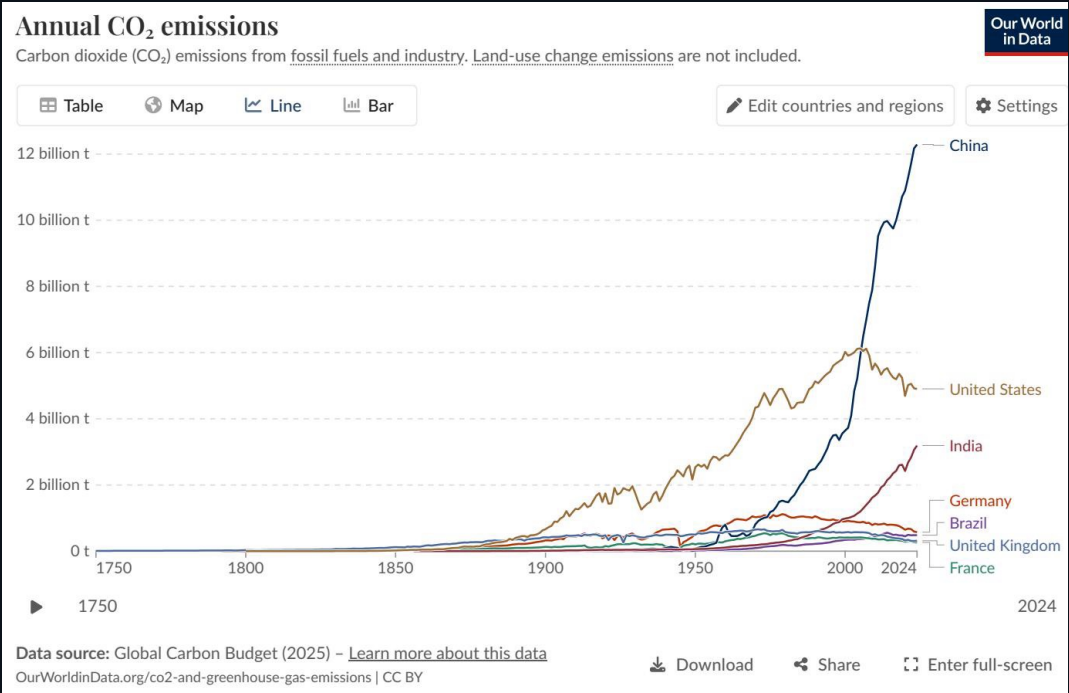
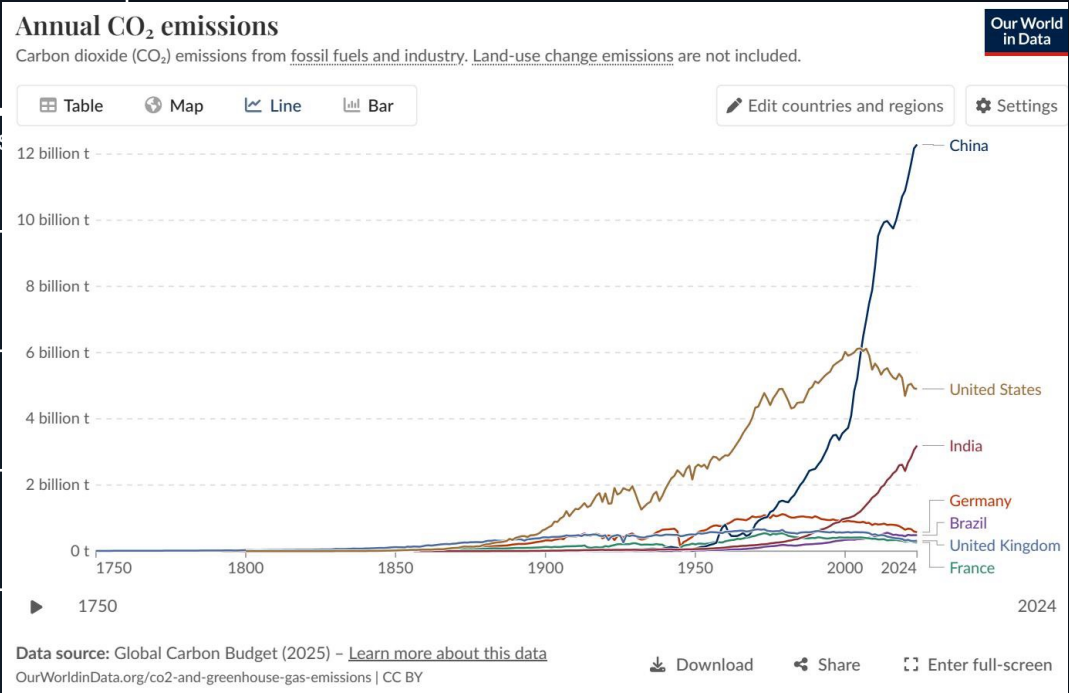


Figure reference in brief: emissions by country (visual proxy for long-run trajectory).

China Now Emits Over 12 Billion Tonnes — More Than Double the U.S.

Country / Region	Annual CO ₂ (Bt)	Share	Trend
China	~12.5	~27%	Slight increase
United States	~5.0	~14%	Declining
India	~3.0	~8%	Rising
EU-27	~2.7	~7%	Declining
Germany	~0.6	~1.7%	Declining
United Kingdom	~0.3	~0.9%	Declining
Brazil	~0.5	~1.4%	Roughly stable
France	~0.3	~0.8%	Declining

Source: Global Carbon Budget 2025 (via Our World in Data)



Figure_3: Annual CO₂ emissions by country

Asia Now Accounts for ~50% of Global Emissions; US + Europe Bel

1900 (share)

Europe + US: >90%
(Other regions: <10%)

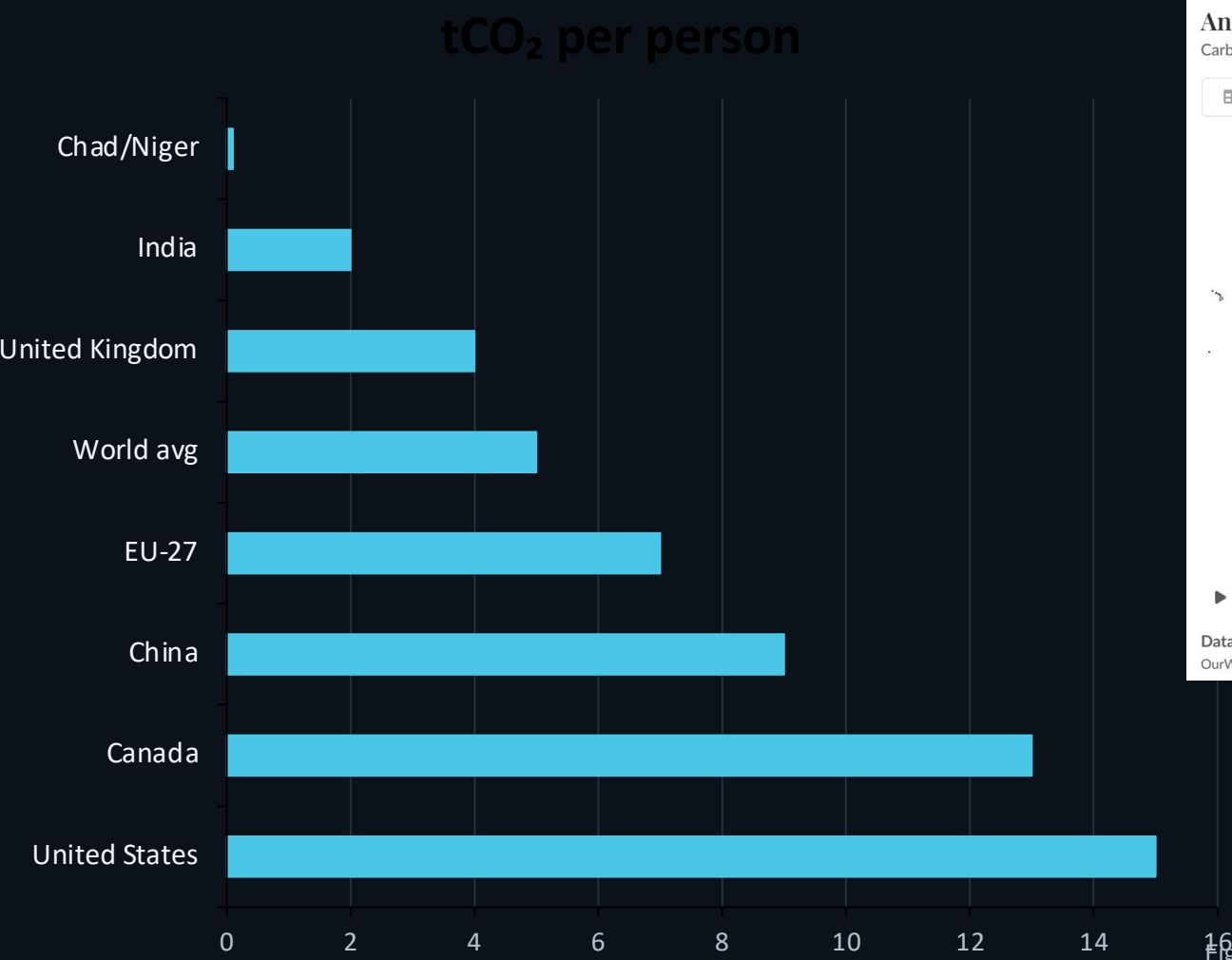
2024 (share)

Asia: ~50%
US + Europe: <33%
Africa: ~3–4%
South America: ~3–4%

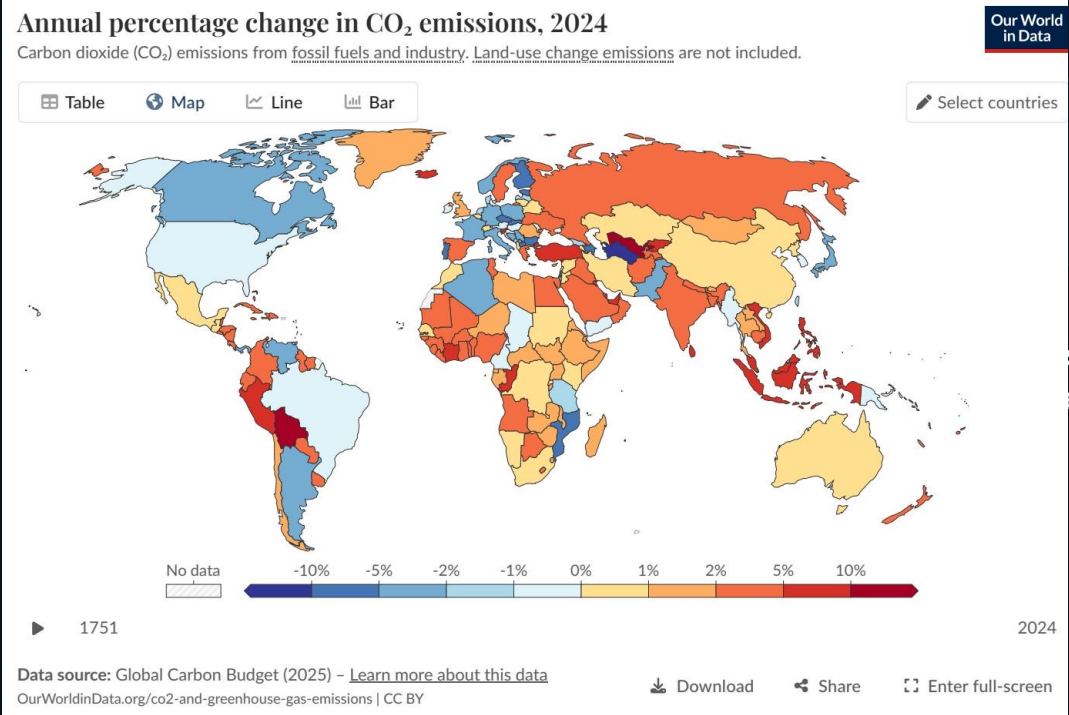
Visual concept: stacked area (regional shares over time) or 1900 vs 2024 side-by-side share blocks.

Source: Global Carbon Budget 2025 (via Our World in Data)

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



Source: Global Carbon Budget 2025 (via Our World in Data)



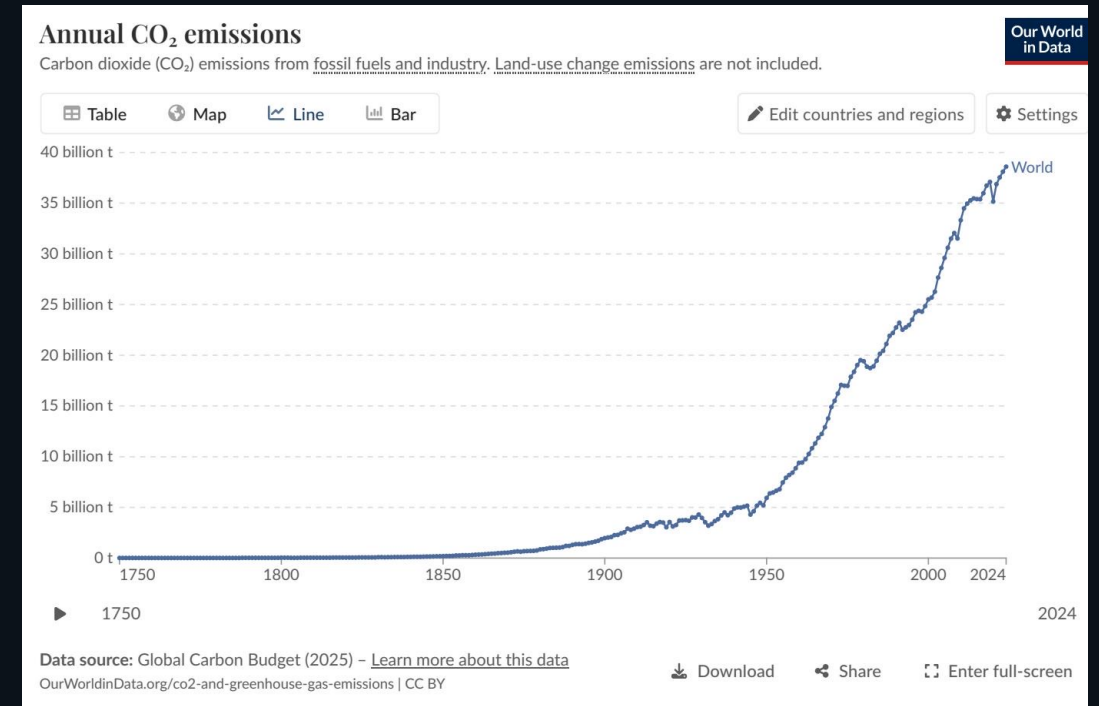
Figure_1: Per-capita CO₂ emissions over time (reference)

2024 Snapshot: US and Europe Cut Emissions While Russia and SE

US and much of Europe: declines of roughly -1% to -5% (blue).

Russia, parts of Southeast Asia, and several African nations: $+2\%$ to $+10\%$ or more (orange/red).

South America: mixed outcomes; one or two countries show very large increases ($>10\%$).



Figure_4: Annual % change in CO₂ emissions (2024)

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Similar living standards can produce different per-capita CO₂ outcomes.

Higher nuclear and renewable shares correlate with lower per-capita emissions.

Higher fossil fuel reliance tends to raise per-capita emissions.

Country group	Energy sources (high-level)	Per-capita CO ₂
France / UK	Higher nuclear + renewables	~4 t/person
Germany / Netherlands	Higher fossil fuel share	Higher than France/UK (per brief)

Key Takeaways: Emissions Are Still Rising, But the Map of Respons

Global CO₂ emissions exceed 35 Bt/yr and have not yet peaked.

China emits ~12.5 Bt (~27%) — more than double the US (~5 Bt).

Regional center shifted: Europe+US >90% (1900) → Asia ~50% today; Europe+US <33%.

Per-capita inequality remains extreme: ~150× gap (US/Australia ~15 t vs Chad/Niger ~0.1 t).

Energy mix and policy choices shape per-capita outcomes — not prosperity alone.